Yongdong Yeo

https://yongdongyeo.github.io | yongdong@snu.ac.kr | +82 10-4114-9921

Summary

I am a Ph.D. student advised by **Prof. Jung Hee Cheon** at the Department of Mathematical Science, Seoul National University. My research area includes various topics of **Homomorphic Encryption**, including its applications and related protocols.

Education

2022 - present Research Area: Homomorphic Encryption and Cryptography
2019 - present Ph.D. student at **Department of Mathematical Science**, **Seoul National University** (GPA: 3.57/4.3)
2014 - 2019 B.S. at **Department of Mathematics, Konkuk University**

Publications

Conference & Journal

[1] Jung Hee Cheon, Hyeongmin Choe, Seunghong Kim, and Yongdong Yeo. "Multi-Party Homomorphic Encryption with Dynamicity and Ciphertext Reusability". In: *IACR Communications in Cryptology* 3.1 (May 4, 2026). ISSN: 3006-5496. DOI: 10.62056/a0qj89krz.

[2] Keewoo Lee and Yongdong Yeo. *SophOMR: Improved Oblivious Message Retrieval from SIMD-Aware Homomorphic Compression*. Cryptology ePrint Archive, Paper 2024/1814. To appear in USENIX Security. 2026. URL: https://eprint.iacr.org/2024/1814.

[3] Jung Hee Cheon, Minsik Kang, Taeseong Kim, Junyoung Jung, and Yongdong Yeo. "Batch Inference on Deep Convolutional Neural Networks With Fully Homomorphic Encryption Using Channel-By-Channel Convolutions". In: *IEEE Transactions on Dependable and Secure Computing* 22.2 (2025), pp. 1674–1685. DOI: 10.1109/TDSC.2024.3448406.

[4] Jihwan Kim, Jung Hee Cheon, and Yongdong Yeo. "OverModRaise: Reducing Modulus Consumption of CKKS Bootstrapping". In: *IACR Communications in Cryptology* 2.3 (Oct. 6, 2025). ISSN: 3006-5496. DOI: 10.62056/a3n5qjp10.

Preprints

[1] Jung Hee Cheon, Keewoo Lee, Jai Hyun Park, and Yongdong Yeo. *SIMD-Aware Homomorphic Compression and Application to Private Database Query*. 2024. arXiv: 2408.17063 [cs.CR]. URL: https://arxiv.org/abs/2408.17063.

Honors & Awards

(Korea) National Cryptography Contest

2025,	Encouragement Award, $1500	SophOMR: Improved Oblivious Message Retrieval from SIMD-Aware Homomorphic Compression
2024,	Encouragement Award, $1500	Reusable Dynamic Multi-Party Homomorphic Encryption
2024,	Special Award, $500	OverModRaise: Reducing Modulus Consumption of CKKS Bootstrapping
2023,	Special Award, $500	Private Database Queries with SIMD-Aware Homomorphic Compression
2023,	Special Award, $500	Batch Inference on Deep Convolutional Neural Networks with Fully Homomorphic Encryption Using Channel-By-Channel Convolutions

Invited Talks & Presentations

2026 Aug.,	USENIX Security 2026:	SophOMR: Improved oblivious message retrieval from SIMD-aware homomorphic compression
2026 Jun.,	PACOH KIAS Workshop:	SophOMR: Improved oblivious message retrieval from SIMD-aware homomorphic compression
2025 spring,	KMS Spring Meeting:	SophOMR: Improved oblivious message retrieval from SIMD-aware homomorphic compression
2024 fall,	KMS Annual Meeting:	Reusable Dynamic Multi-Party Homomorphic Encryption
2024 Aug.,	MPC & SNARK Workshop:	SophOMR: Improved oblivious message retrieval from SIMD-aware homomorphic compression
2023 Dec.,	Dept. of Math at Konkuk University	Homomorphic Encryption and Its Application to Machine Learning
2023 fall,	KMS Annual Meeting:	Private database query with SIMD-aware homomorphic compression

Teaching

To be Updated.

Languages

Korean	Native language. Fluent.
English	2nd language. Proficient.
Japanese	3rd language. Conversational.
Programming language:	C++, Go, Python, MatLab. Experience with AI-assisted tools.